



NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

STR

LAB #1 – class 0/5

SW INSTALLATION &

INTERACTION WITH THE “PHYSICAL” SYSTEM

2025 - 2026

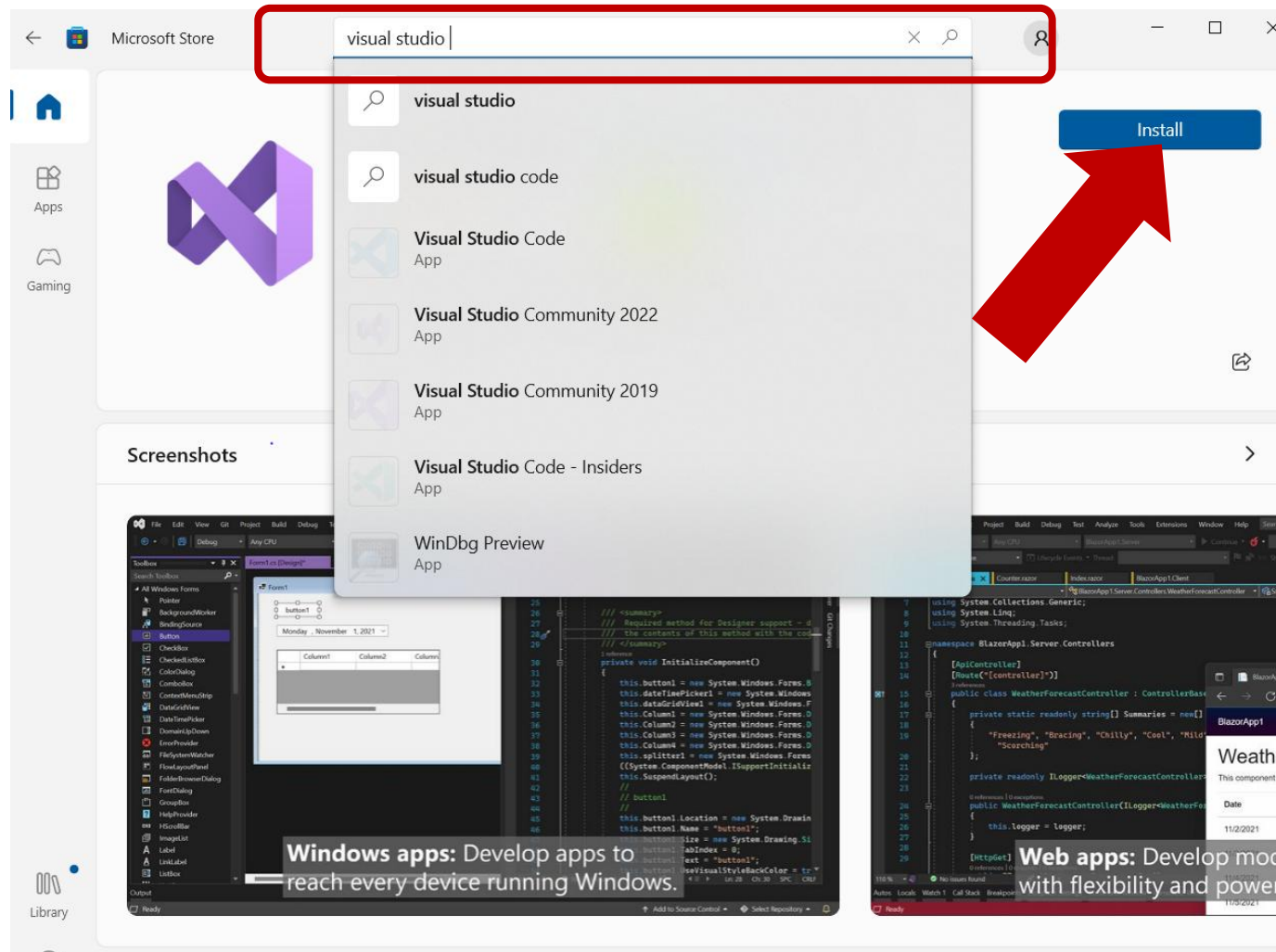
Software installation & configuration

- ☐ Software installation
- ☐ Creation and configuration of new project for windows 32 console application
- ☐ Adding components to the project

INSTALLATION OF VISUAL STUDIO



Install Visual Studio Community 2022 using Microsoft Store or <https://visualstudio.microsoft.com/downloads/>



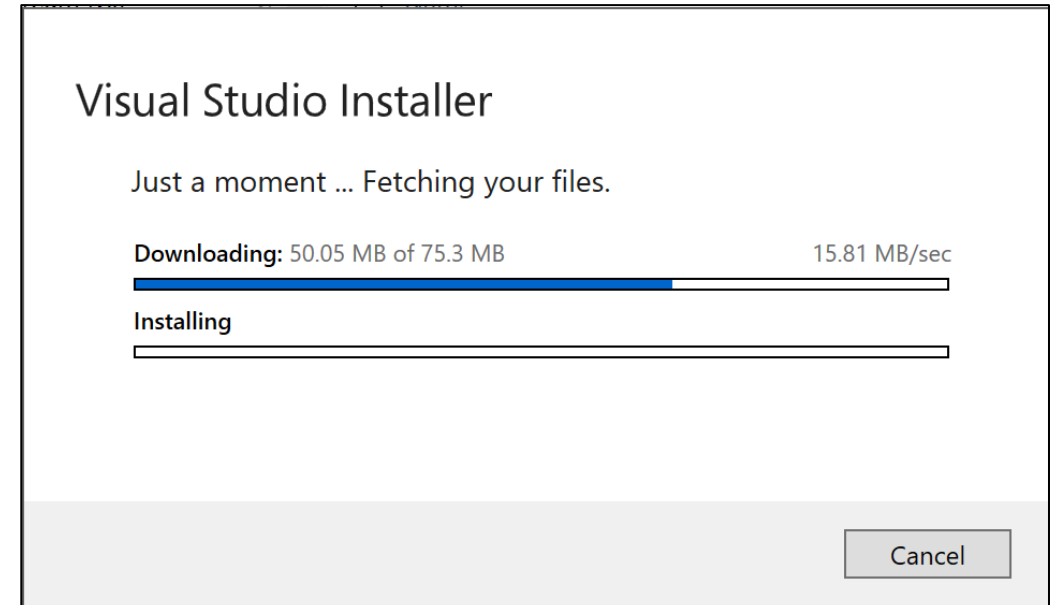
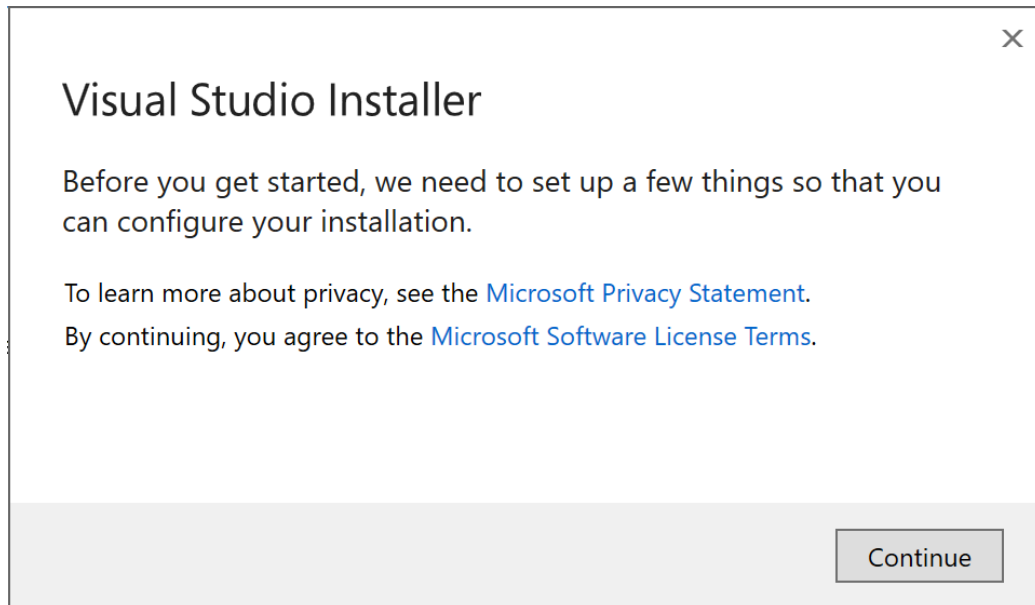
Recommended version:

- Visual Studio 2022
- Community edition

INSTALLATION OF VISUAL STUDIO



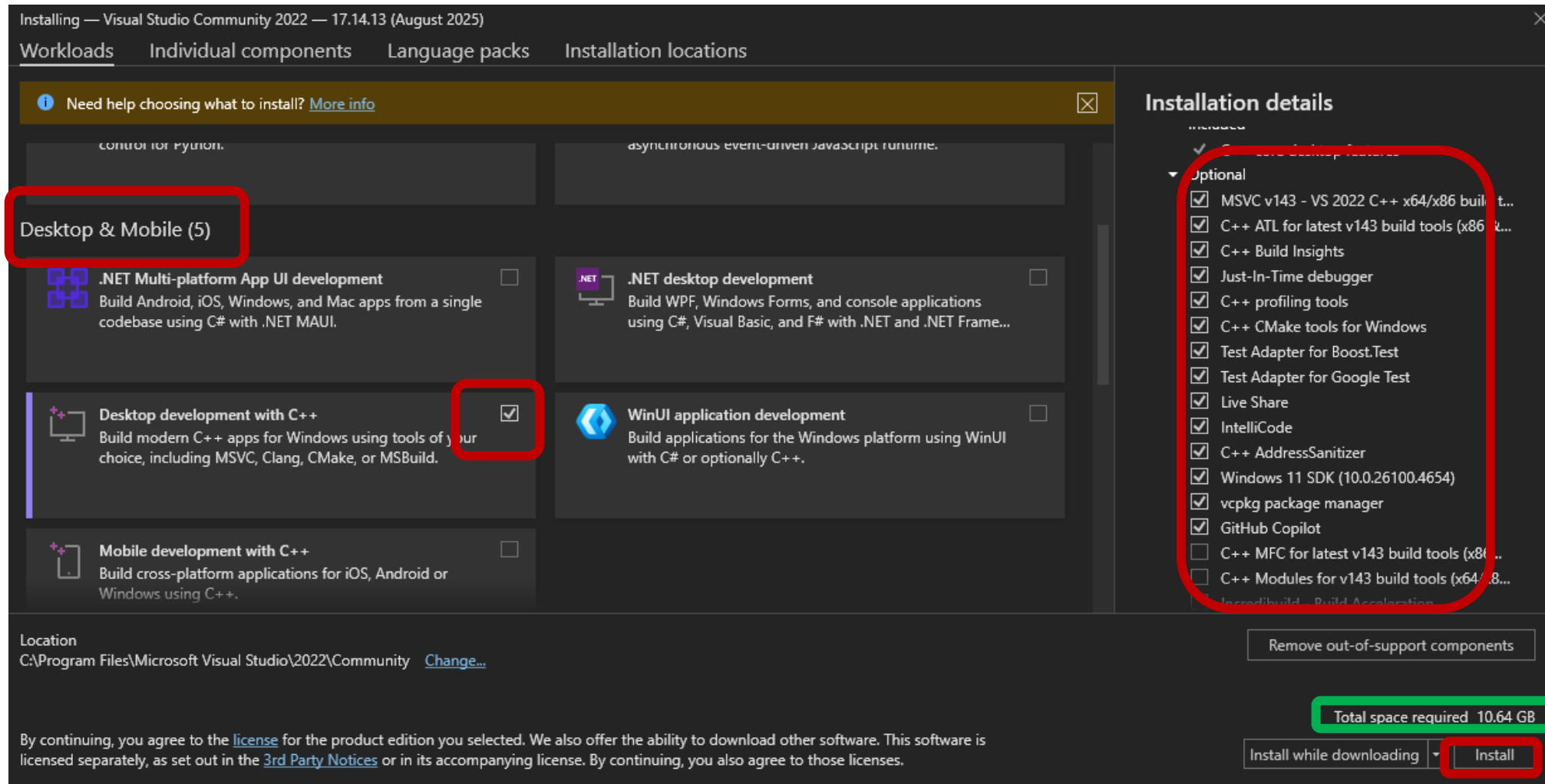
Run installation file



INSTALLATION OF VISUAL STUDIO

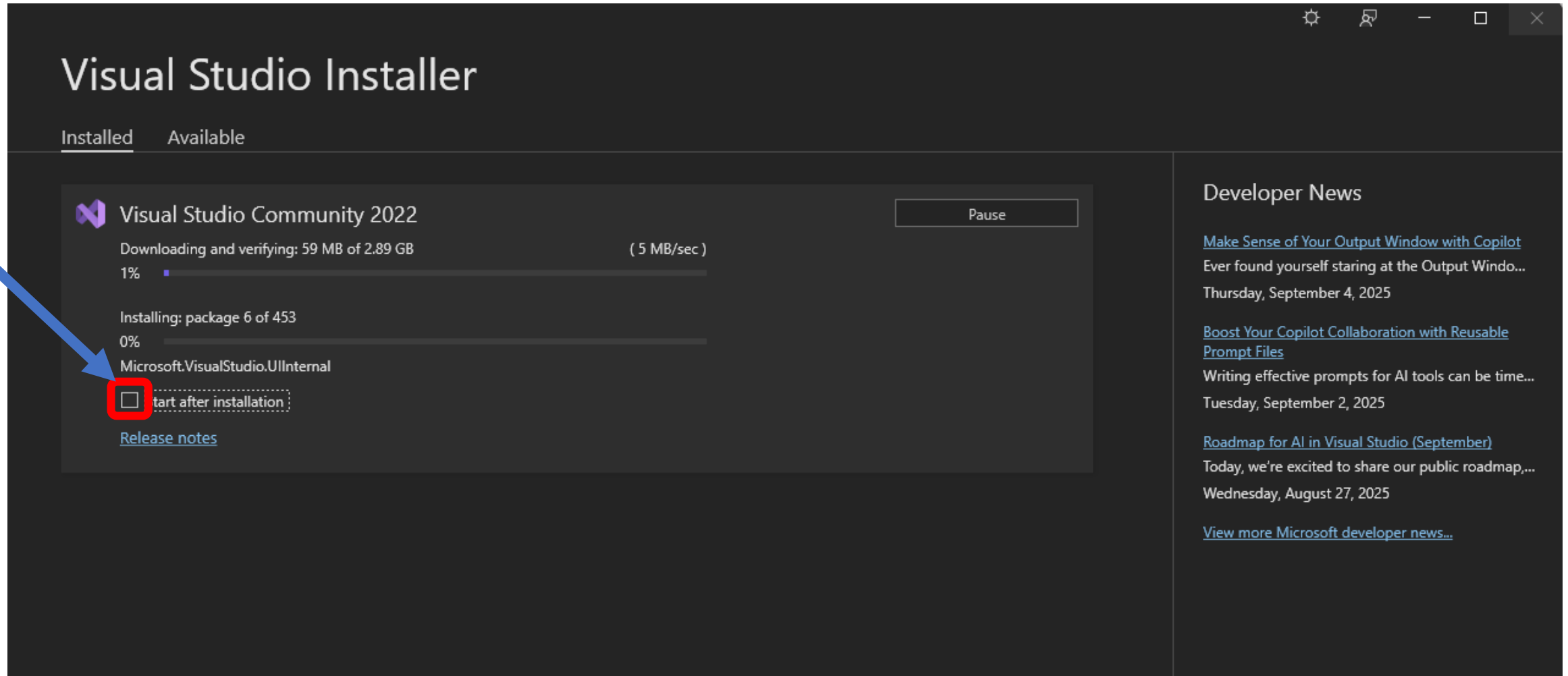


Example for VS Community 2022 features configuration



- To avoid taking unnecessary disk space, select at least these features.
- Additional features maybe installed latter on if necessary.

INSTALLATION OF VISUAL STUDIO



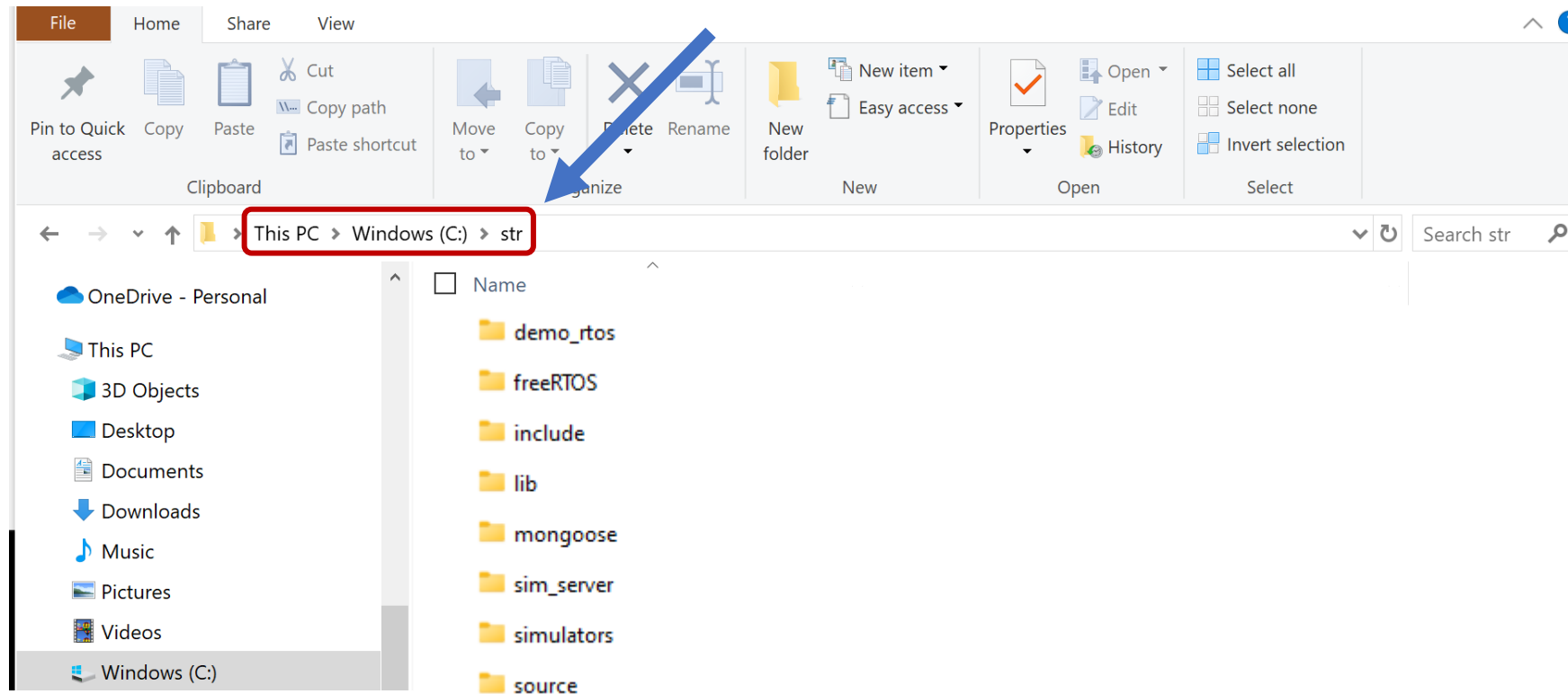
Wait till installation is complete

Do not open the Visual Studio after finishing the installation

LABWORK1 PROJECT PACKAGE



- ❑ Copy the material for Lab work from CLIP
- ❑ Extract **STR_TP1.zip** into **c:\str**, as shown in the picture below.



(*) please comply with directory, as it allows faster evaluation of your lab works.

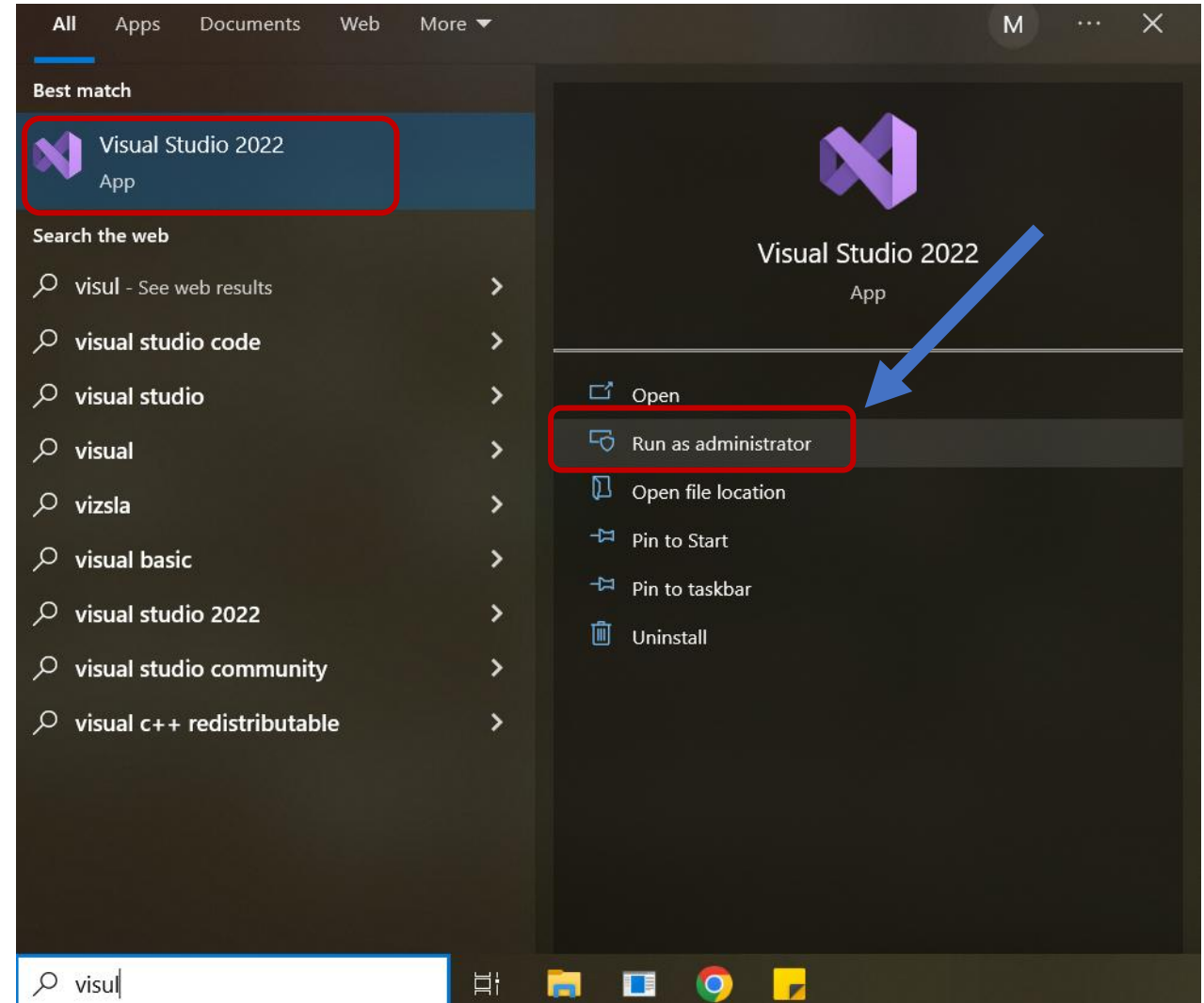
RUN VISUAL STUDIO 2022

Run VS22

- Find Visual Studio 2022 in your startup menu
- Right click with the mouse
- Select “Run as administrator” (*)

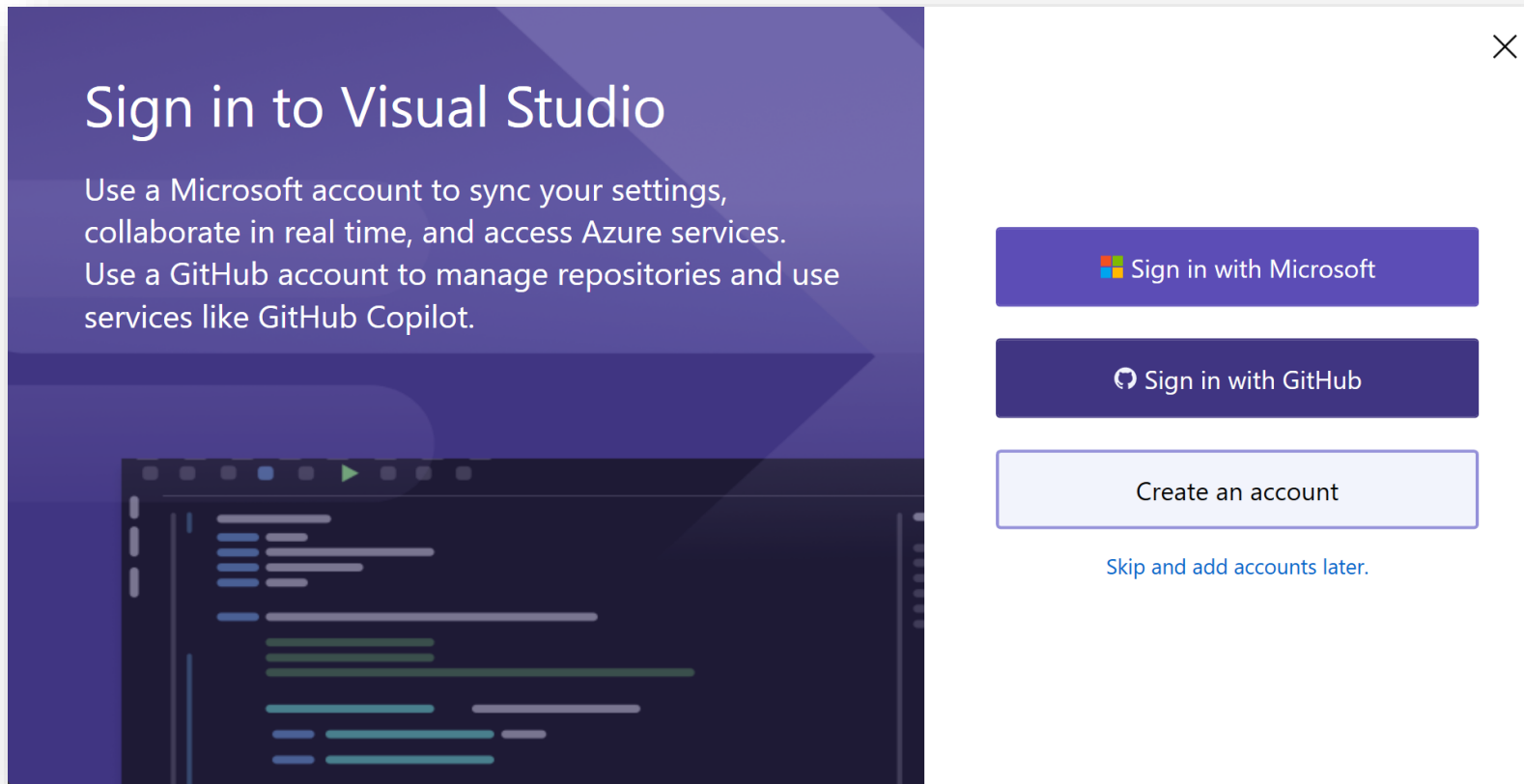
(*) This ensures that our project will obtain more privileges from the Operating System (Windows), ensuring reliable execution in a real time context.

This is important!



❑ Signing in

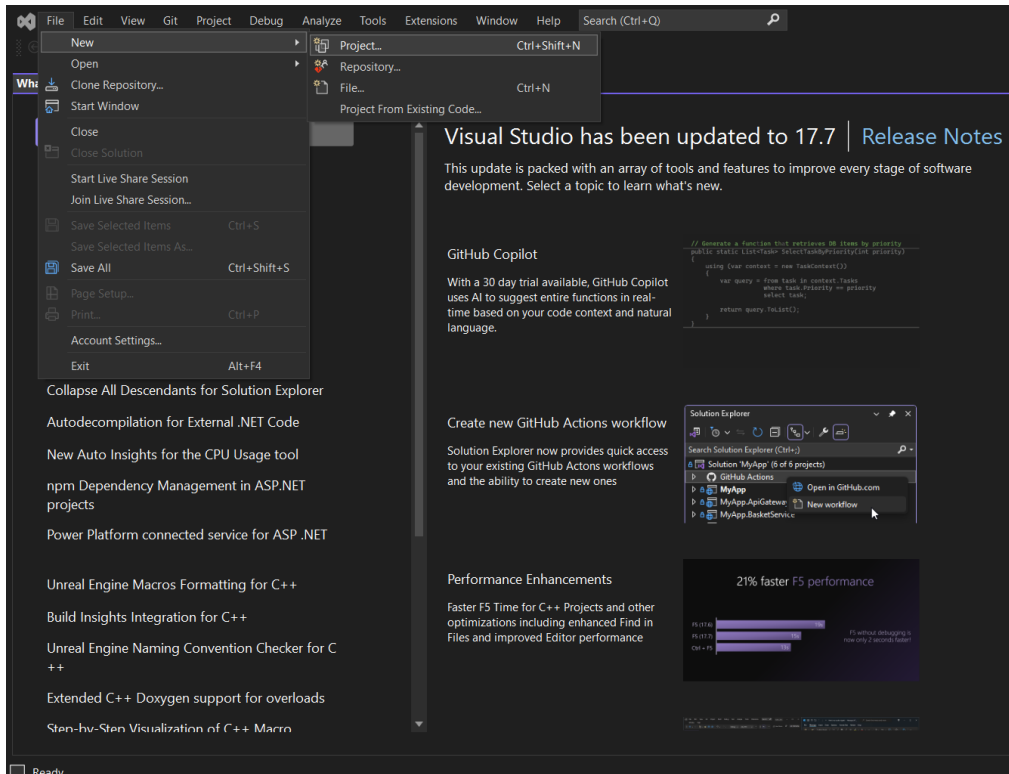
- “Sign in” with your faculty credentials (student login and password) or select “skip and add account later”.
- For the credentials, note that it is necessary to provide them as student_id@fct.unl.pt **(without “campus”)**.



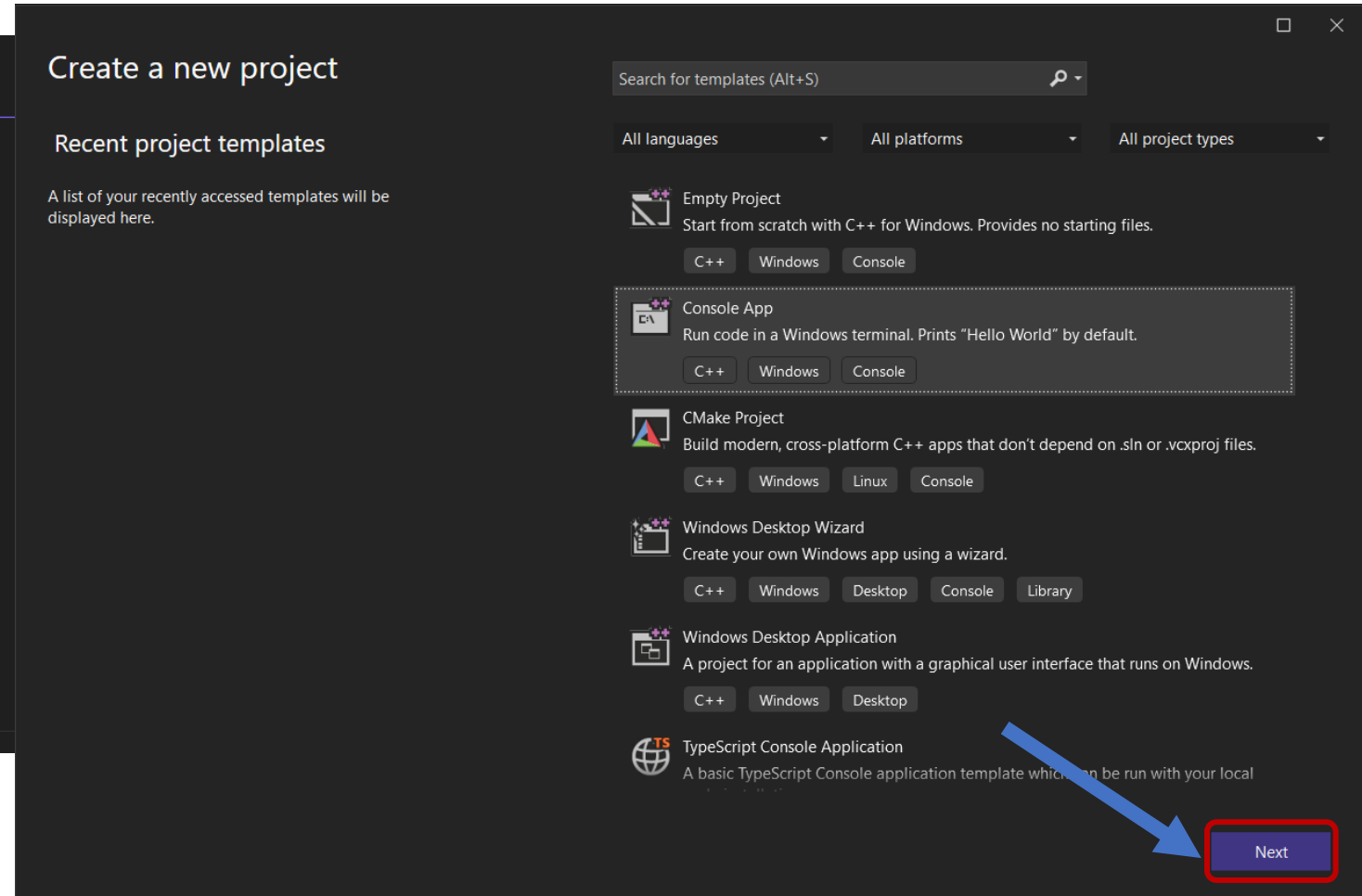
CREATE LABWORK1 PROJECT



Click File > New > Project



Select **Console App** project



CREATE LABWORK1 PROJECT



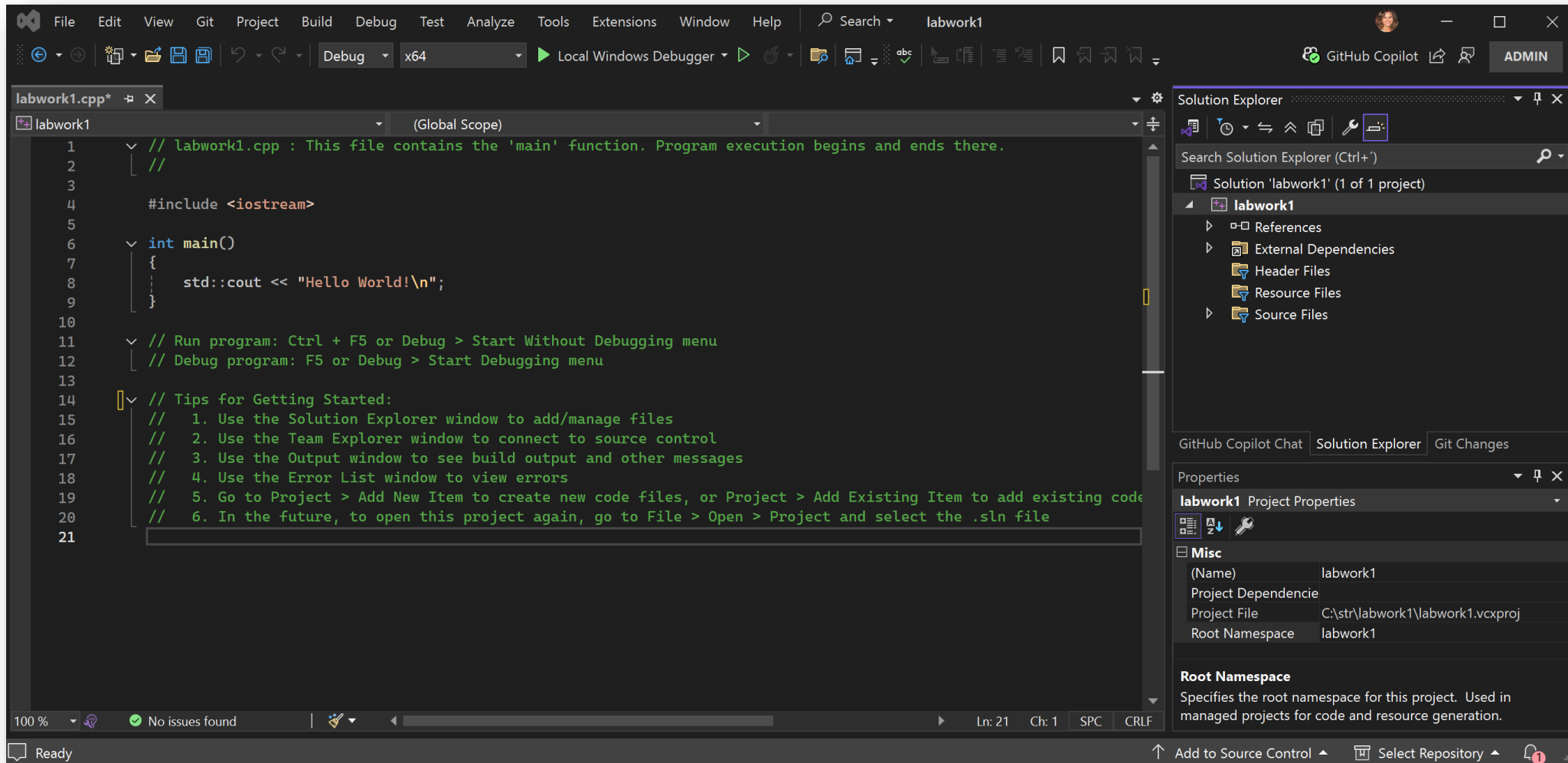
- ☐ Project name: **labwork1**
- ☐ Location: **c:\str**
- ☐ Click on the checkbox.

A screenshot of the Visual Studio "Configure your new project" dialog box. The dialog has a dark theme. At the top, it says "Configure your new project". Below that are three tabs: "Console App" (selected), "C++", "Windows", and "Console". The "Project name" field contains "labwork1". The "Location" field contains "C:\str\". The "Solution name" field contains "labwork1". A checkbox labeled "Place solution and project in the same directory" is checked. Below these fields, it says "Project will be created in 'C:\str\labwork1\'". At the bottom right, there are two buttons: "Back" and "Create". A large blue arrow points down to the "Create" button, which is also highlighted with a red rectangle. The "Project name", "Location", and "Create" button are also highlighted with red rectangles.

CREATE LABWORK1 PROJECT



□ Your console App project



CONFIGURE PROJECT PROPERTIES



□ Right-click labwork1 → Select Properties

The screenshot shows the Visual Studio IDE with a C# project named 'labwork1'. The Solution Explorer on the right shows the project structure. A right-click context menu is open over the 'labwork1' project. The 'Properties' option at the bottom of the menu is highlighted with a red box and a blue arrow. The menu includes options like Rebuild, Clean, View, Analyze and Code Cleanup, Project Only, Retarget Projects, Collapse All Descendants, Scope to This, New Solution Explorer View, Run Build Insights on Selected Files, Build Dependencies, Add, Class Wizard..., Manage NuGet Packages..., Configure Startup Projects..., Set as Startup Project, Debug, Cut, Paste, Remove, Rename, Unload Project, Load Direct Dependencies, Load Entire Dependency Tree, Rescan Solution, Display Browsing Database Errors, Clear Browsing Database Errors, Open Folder in File Explorer, Open in Terminal, and Properties.

```
1 // labwork1.cpp : This file contains the 'main' function. Program execution begins and ends there.
2
3
4 #include <iostream>
5
6 int main()
7 {
8     std::cout << "Hello World!\n";
9 }
10
11 // Run program: Ctrl + F5 or Debug > Start Without Debugging menu
12 // Debug program: F5 or Debug > Start Debugging menu
13
14 // Tips for Getting Started:
15 // 1. Use the Solution Explorer window to add/manage files
16 // 2. Use the Team Explorer window to connect to source control
17 // 3. Use the Output window to see build output and other messages
18 // 4. Use the Error List window to view errors
19 // 5. Go to Project > Add New Item to create new code files, or Project > Add Existing Item to add existing code
20 // 6. In the future, to open this project again, go to File > Open > Project and select the .sln file
21
```

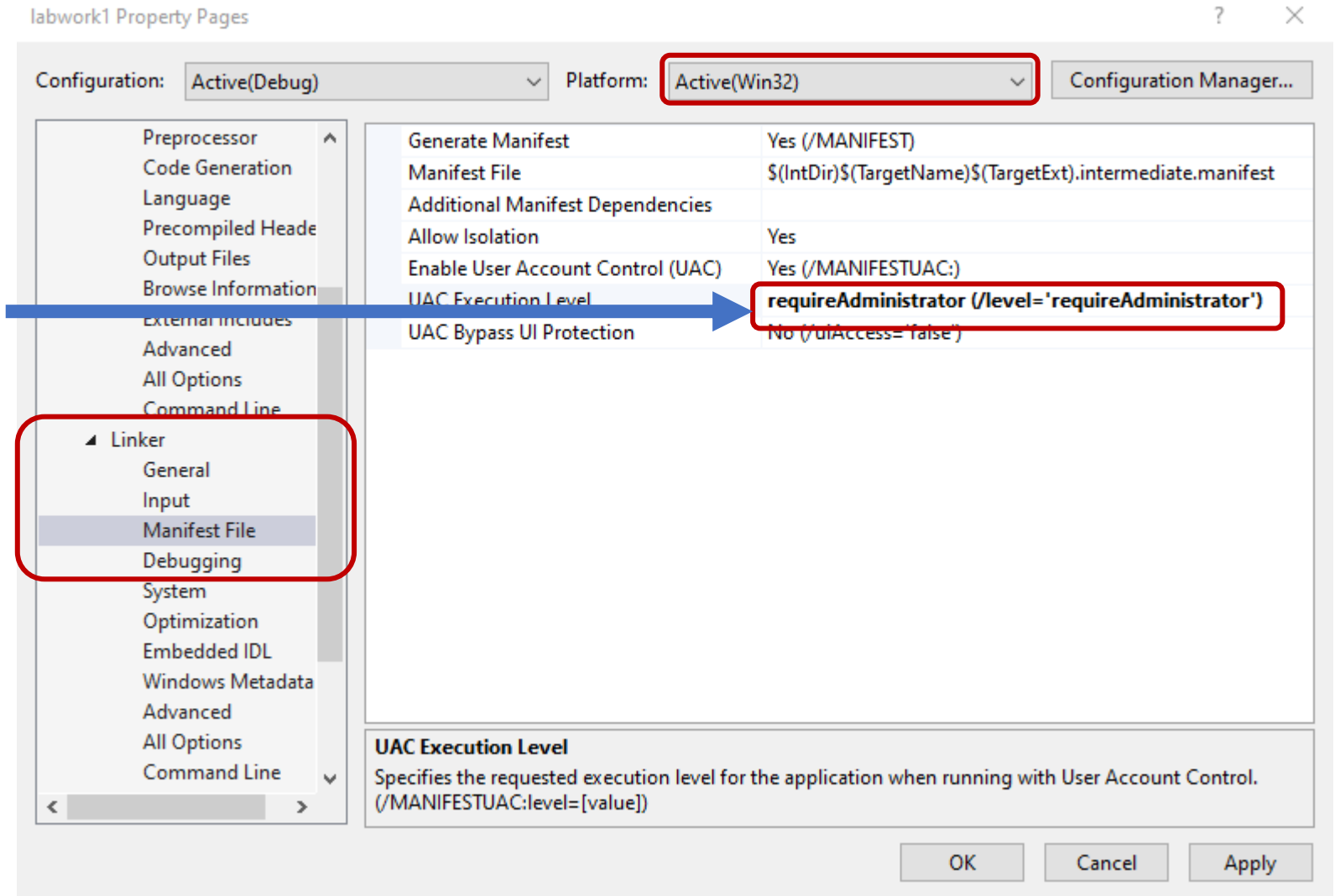

CONFIGURE PROJECT PROPERTIES



- ☐ Select Linker -> Manifest File
- ☐ Change UAC Execution level to “requireAdministrator” (*)

- ☐ Click **Apply + OK**

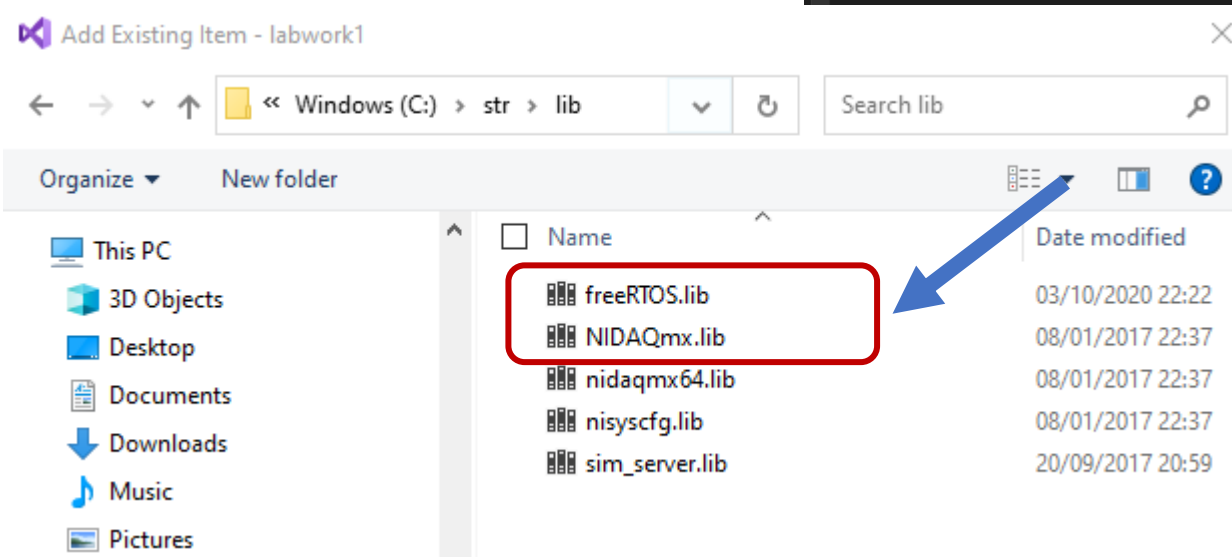
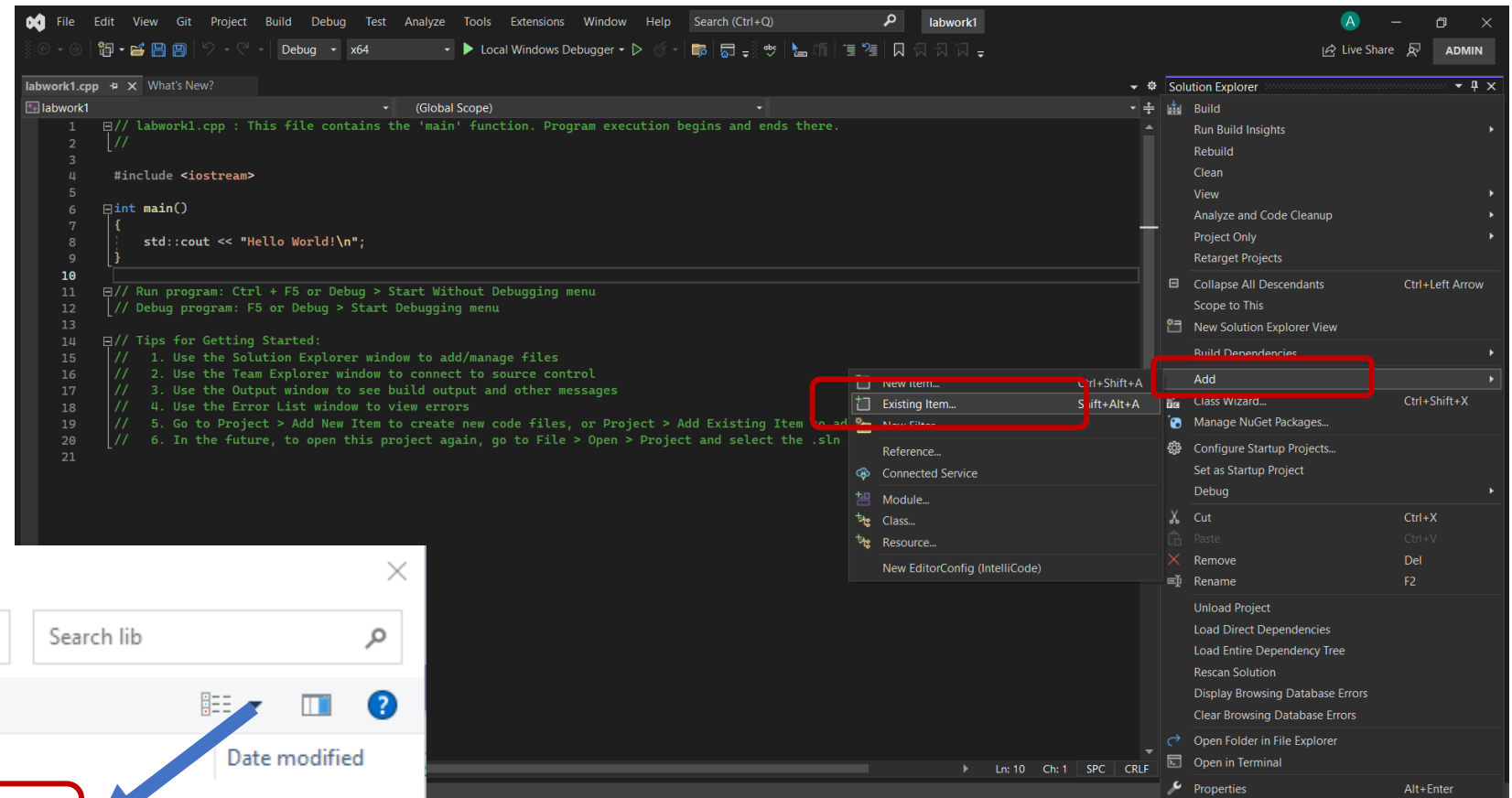
(*) As mentioned before, your application will receive more privileges from Windows O.S., necessary for adequate real-time behavior.



CONFIGURE PROJECT PROPERTIES

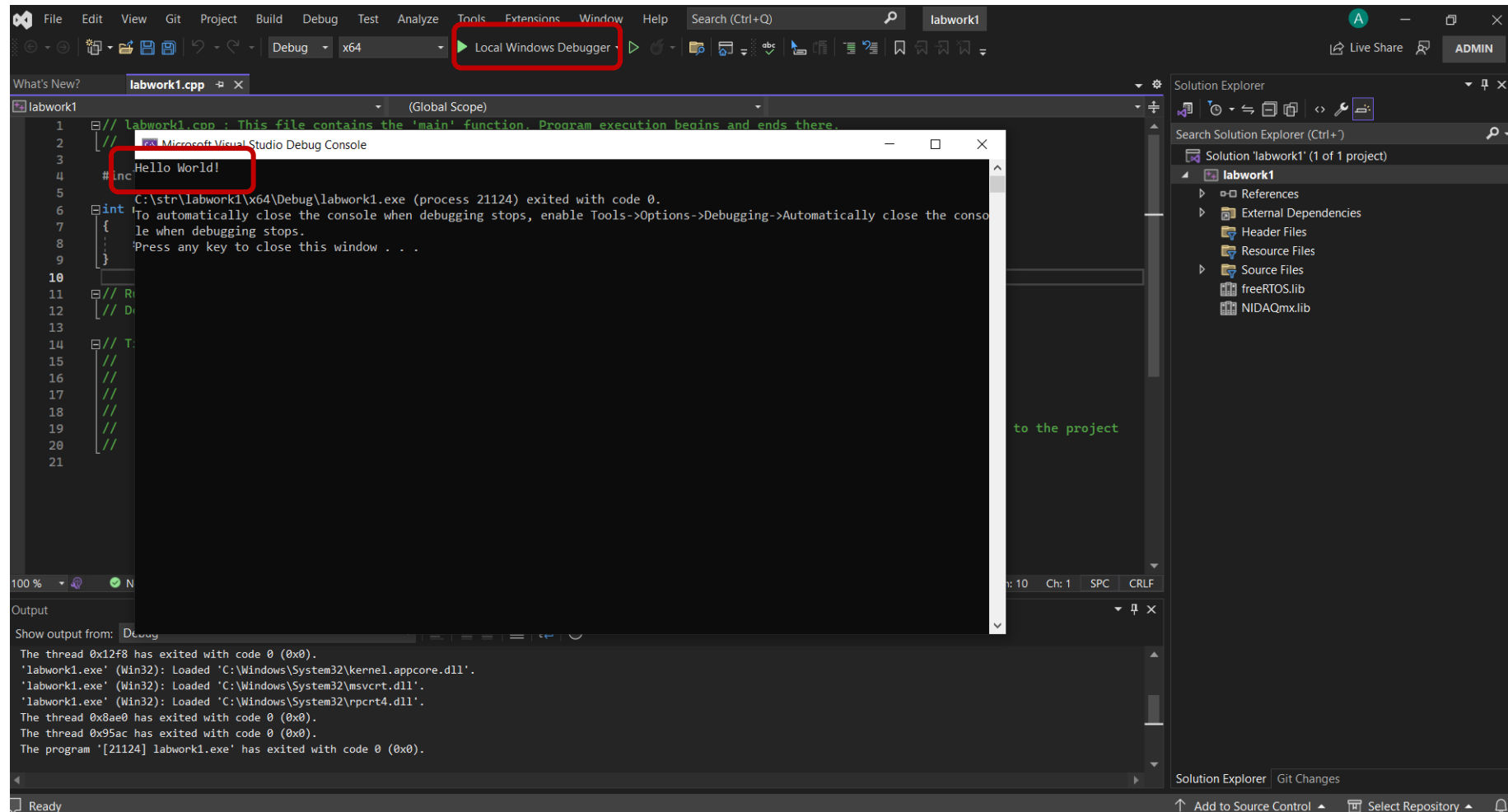


- ❑ Right Click in the project and select: Add > Existing item.
- ❑ In folder c:\str\lib, select the files below.
- ❑ Click Add button.



RUN PROJECT

- ❑ Execute project (Click “Local Windows Debugger” or press F5).
- ❑ Check if it runs



AWESOME! YOU'RE ALL SET TO BEGIN YOUR LAB 1 JOURNEY IN REAL-TIME SYSTEMS!



RELAX, EXPLORE, HAVE FUN, AND ENJOY THE RIDE!



KEEP UP THE
GOOD
WORK!